

Cellular Networks

Overview

1. Cell Structure/Geometry
2. Cellular Frequency Reuse
3. Cellular Handoff
4. Cellular System Capacity
5. Overview of Cellular Generations: 1G □ 2G □ 3G □ LTE/4G □ 5G

Wide Area Networking

- ❑ Bluetooth is good to exchange short messages between two devices located close to each other ($\sim 10\text{m}$)
- ❑ WiFi is good mainly within a home/building ($\sim 20\text{-}50\text{m}$)
- ❑ How about wide area networking (many kilometers)?
- ❑ Cellular networking addresses wide area; much more complex and expensive



Bluetooth



WiFi

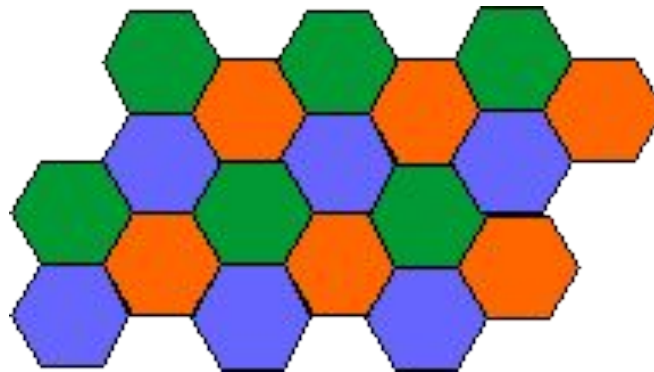


Cellular

Cellular Concept

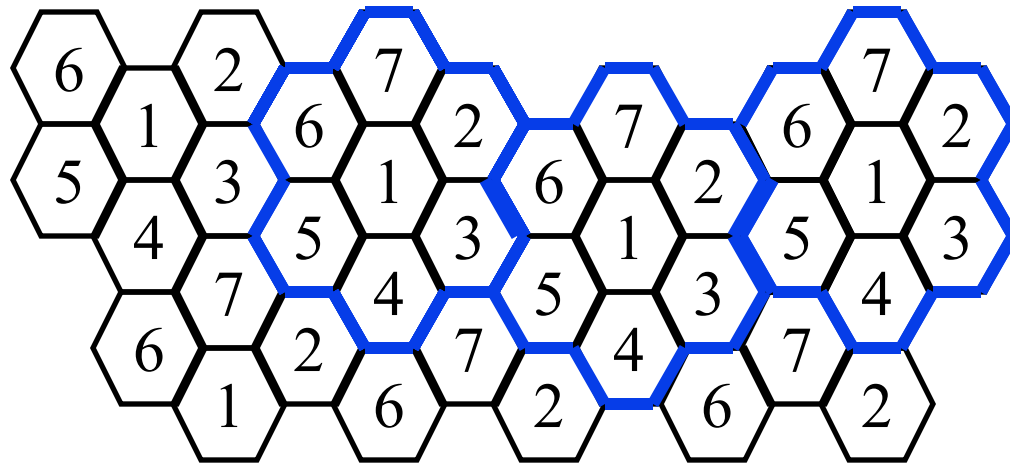
- ❑ First proposed in '70s; commercial services offered early '80s
- ❑ A large geographic service area is divided into many smaller cells; no matter where you are located, you are always within a cell
- ❑ Each cell has a base station to connect users within the cell; all base stations are in turn connected to a central control system
- ❑ Adjacent cells must not use the same frequency to avoid interference
- ❑ The same frequency can be reused by a 'distant' base station increase reuse the spectral resources and increase system capacity

3 frequencies, red,
green, and blue, reused
by distant cells



Cellular Network Beginnings

- ❑ AT&T Bell Labs designed a cellular structure to reuse frequency. No two adjacent cells use the same frequency.
- ❑ 1977: FCC authorized two commercial deployments
 - ❑ Chicago: Illinois Bell
 - ❑ Washington, DC: American Radio telephone Service
 - ❑ Both services started 1983



Ref: P. Bedell, "Cellular Networks: Design and Operation, A real World Perspective," Outskirts Press, 2014, ISBN:9781478732082

Initial Cellular System in US

- ❑ US was divided into
 - ❑ 306 metropolitan service areas (MSAs)
75% of US population, 20% of area
Densely populated $\Rightarrow$ Small cell size
 - ❑ 428 rural service areas (RSAs)
Less populated $\Rightarrow$ Larger cell size
- ❑ Each area was originally allowed two competing carriers: A, B
 - ❑ Bell (B)
 - ❑ Alternative (A)
- ❑ 832 channel-pairs in each area. 416 pairs per carrier.
45 MHz between transmit and receive frequencies
30 kHz per channel
1:7 Frequency reuse with hexagonal cells
- ❑ Too many applicants $\Rightarrow$ FCC started a lottery system
- ❑ At least one system in every market by 1990

Cell Sites

- ❑ On towers, roof tops, water tanks, utility poles, ...
 - ❑ Good source of income for utility companies, cities, schools, churches, hotels, ...
 - ❑ With a base station for electronics
 - ❑ NIMBY (Not in my back yard)
 - ⇒ Mostly hidden, shared towers



Cells on Wheels (CoWs)

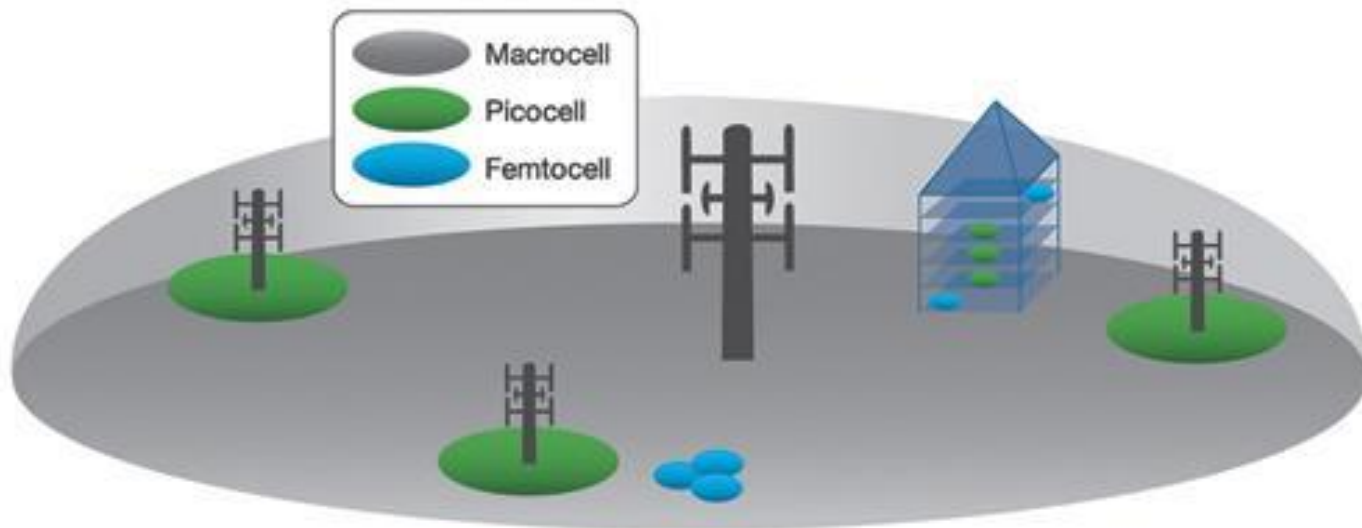
- ❑ Used for temporary surge in traffic, e.g., games, fares, ...



How large are the cells?

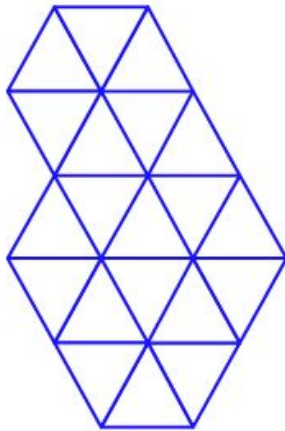
Macro, Micro, Pico, Femto Cells

- ❑ Macro: Sections of a city, more than 1 km radius
- ❑ Micro: Neighborhoods, less than 1 km
- ❑ Pico: Busy public areas: Malls, airports, ..., 200 m
- ❑ Femto: Inside a home, 10 m



Cell Geometry

- ❑ Although there is no regular cell geometry in practice due to natural obstacles to radio propagations, a model is required for planning and evaluation purposes
- ❑ Simple model: All cells have identical geometry and should *tessellate* perfectly to avoid any coverage gaps in the service area
 - ❑ Radio propagation models lead to circular cells, but circles do not tessellate!
- ❑ Three options for tessellation: equilateral **triangle**, **square**, regular **hexagon**
- ❑ Hexagon has the largest area among the three; hence its typical use



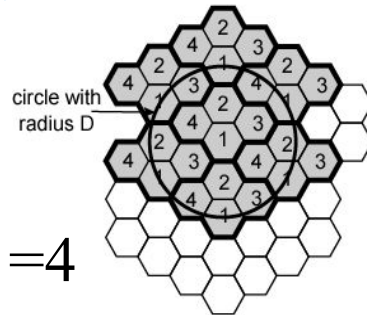
Frequency reuse and clustering

- ❑ Adjacent cells cannot use the same channel due to interference
- ❑ All cells in the service area are grouped into many clusters; the total spectrum is divided into sub-bands that are distributed among the cells within a cluster; the spatial distribution of sub-bands within the cluster should make sure that adjacent cells do not share the same sub-band
- ❑ A cluster of cells together use the entire spectrum
- ❑ By dividing the service area into many clusters, the operator can reuse the allocated spectrum spatially over the entire service area

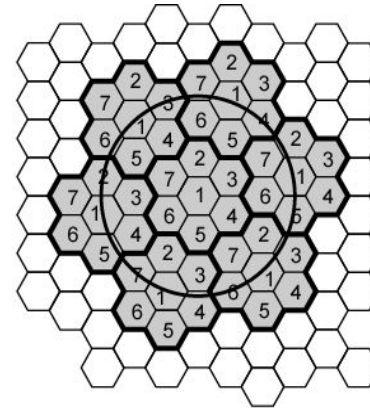
Cluster Size

Clusters are shown
with solid borders;
N represents
cluster size

Cluster Size = 4

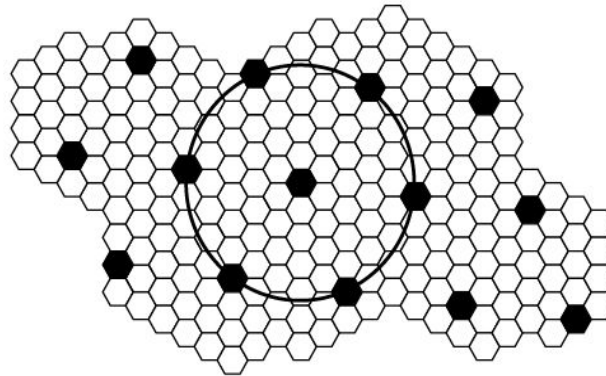


(a) Frequency reuse pattern for $N = 4$



Cluster Size = 7

(b) Frequency reuse pattern for $N = 7$

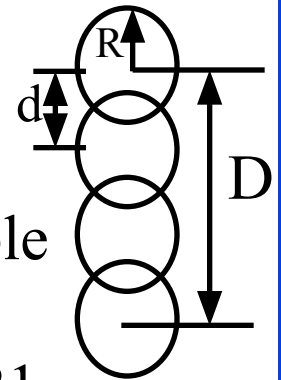


(c) Black cells indicate a frequency reuse for $N = 19$

Cluster Size = 19

Characterizing Frequency Reuse

- ❑ **D** = minimum distance between centers of cells that use the same band of frequencies (called co-channels)
- ❑ **R** = radius of a cell
- ❑ **d** = distance between centers of adjacent cells ($d = R\sqrt{3}$)
 - ❑ $d < 2R$ due to *overlapping* cells
- ❑ **N** = number of cells in repetitious pattern (**Cluster**)
 - ❑ **Frequency Reuse Factor** = $1/N$
 - ❑ Each cell in cluster uses unique band of frequencies
- ❑ For hexagonal cells, following values of N are possible
 - ❑ $N = I^2 + J^2 + (I \times J)$, $I, J = 0, 1, 2, 3, \dots$
- ❑ Possible values of N are 1, 3, 4, 7, 9, 12, 13, 16, 19, 21, ...
- ❑ **Reuse Ratio** = Distance/Radius = $D/R = \sqrt{3N}$
- ❑ $D/d = \sqrt{N}$



Example

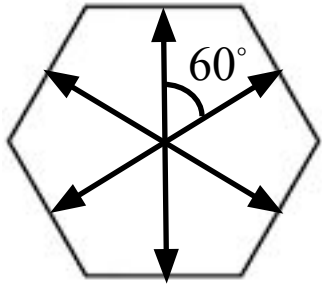
Q. What would be the minimum distance between the centers of two cells with the same band of frequencies if *cell radius* is 1 km and the *reuse factor* is 1/12?

Sol. $R = 1 \text{ km}$, $N = 12$

$$D/R = \sqrt{3N}$$

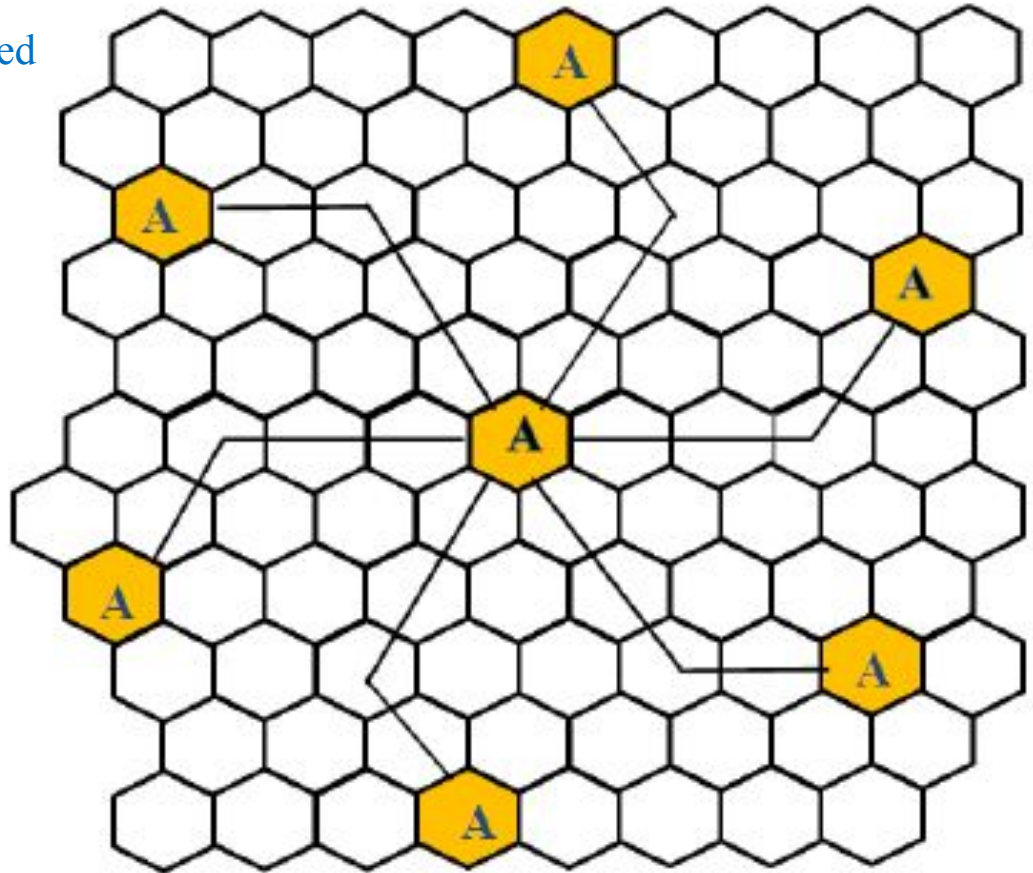
$$\begin{aligned} D &= (3 \times 12)^{1/2} \times 1 \text{ km} \\ &= 6 \text{ km} \end{aligned}$$

Locating Co-channel Cells



6 directions of a hexagon, separated by 60 degrees

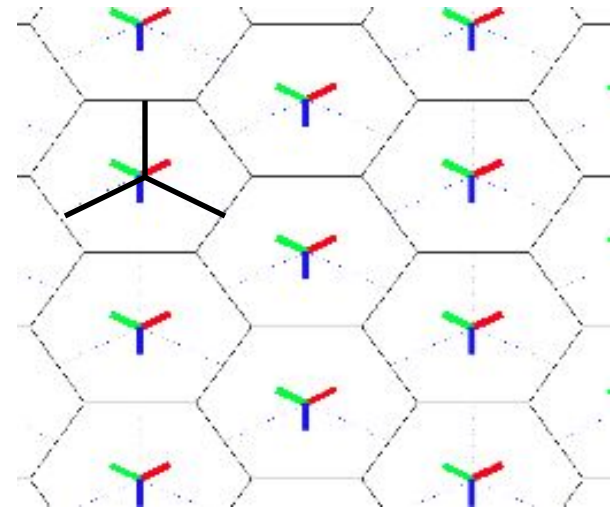
- Move i cells in any direction
- Turn 60° counter-clock and move j cells



$i=3, j=2; N=19$

How to distribute channels among cells within a cluster?

- For simplicity, it is assumed that the total spectrum is divided equally among all cells in the cluster
 - T (total channels), N (cluster size), K (number of channels per cell)
 - $K = T/N$
- Cells are usually divided into sectors; channels allocated to a cell is then further sub-allocated to different sectors according to the load/demand in each sector; spatial allocation of channels to sectors should try to minimize interference/overlap with the adjacent cell sectors

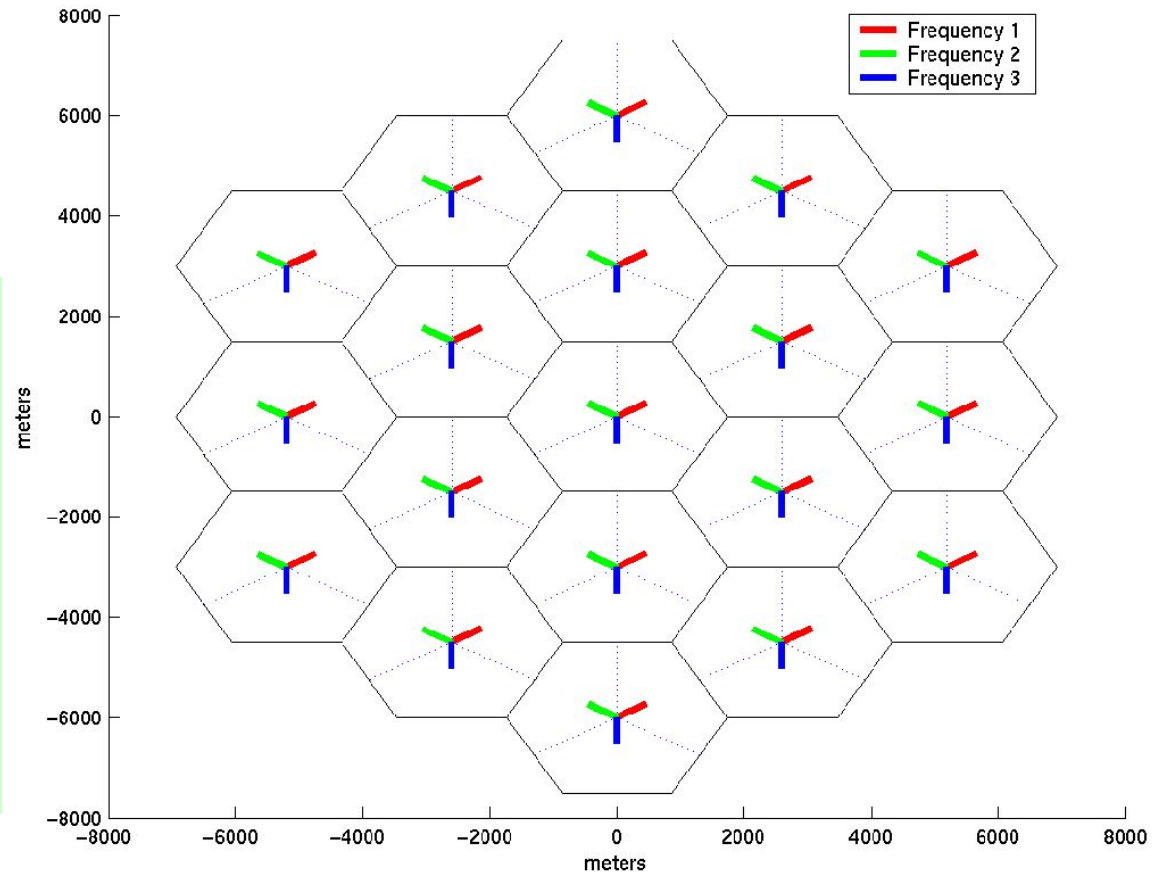


Frequency Reuse Notation

- $N \times S \times K$ frequency reuse pattern
- N = Number of cells per cluster
- S = Number of sectors in a cell
- K = Number of frequency/channel allocations per cell

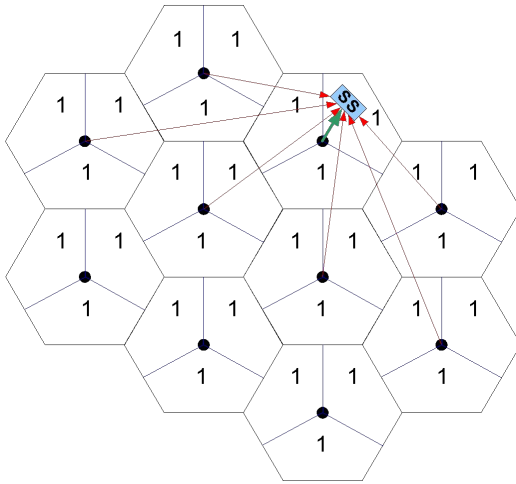
1X3X3

In this case, K is evenly distributed among all sectors. Uneven allocations can address uneven demands in different sectors. $N \times S \times K$ notation does not capture the frequency distribution among sectors.

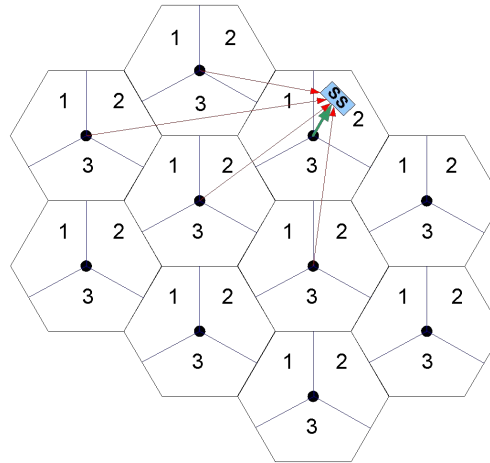


Frequency Reuse Notation (Cont)

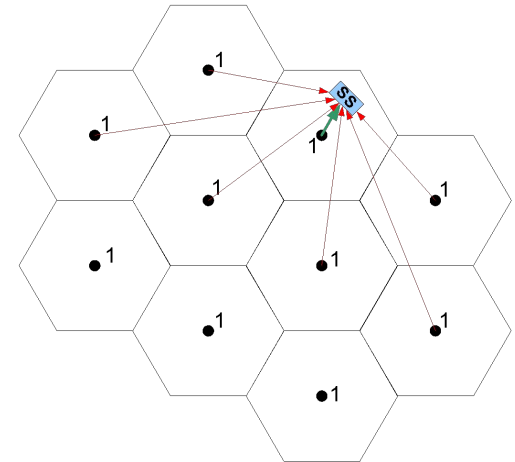
1x3x1



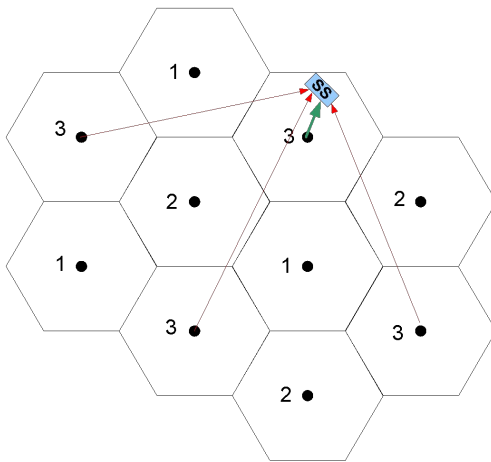
1x3x3



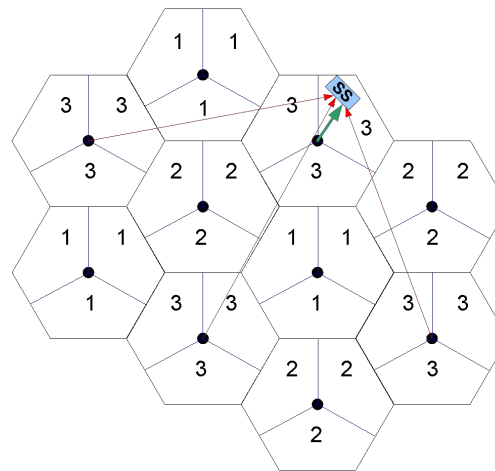
1x1x1



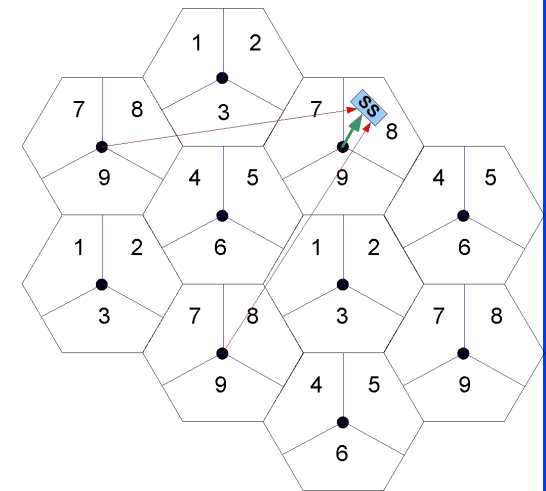
3x1x1



3x3x1

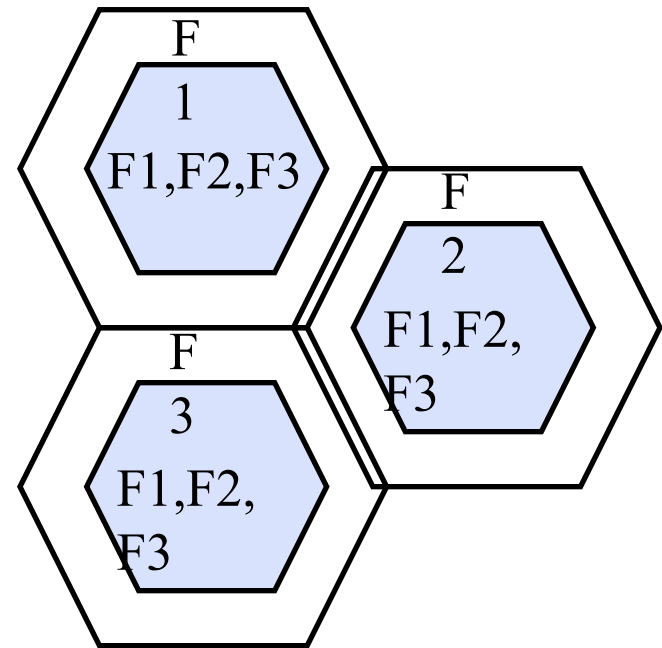


3x3x3



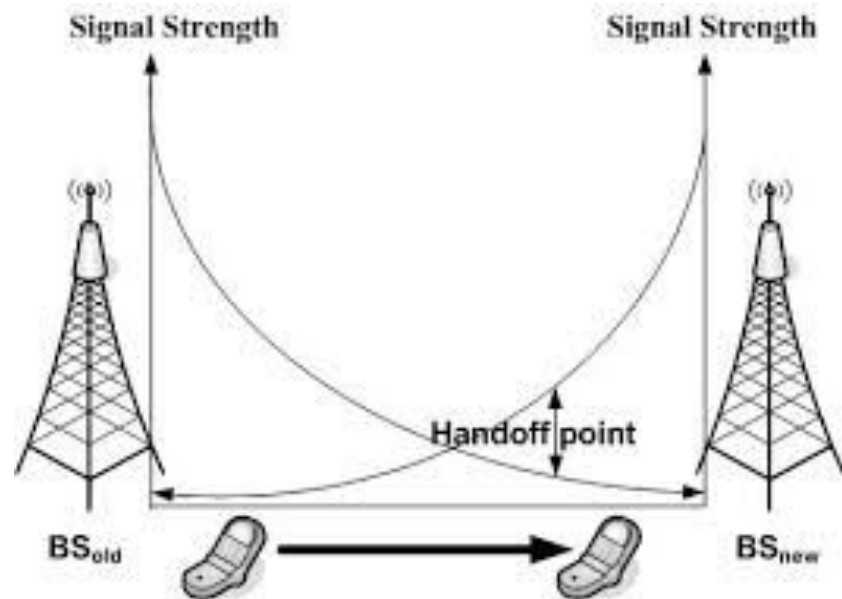
Fractional Frequency Reuse

- ❑ Users close to the BS use all frequencies
- ❑ Users at the cell boundary use only a fraction of available frequencies
- ❑ Border frequencies are designed to avoid interference with adjacent cells;



Handoff

- ❑ User mobility poses challenges for cellular networks; cannot remain connected to the same BS; as the RSS becomes too weak, the mobile device must connect to a new BS with a stronger RSS
- ❑ Disconnecting from one and connecting to a new BS during an on-going session is called handoff



Frequency Allocation for Handoff

- ❑ To handoff successfully, the new BS must have available channels to support the on-going call; otherwise the call will be dropped
- ❑ Dropping an ongoing call is worse than rejecting a new call
- ❑ BSs therefore usually reserve some channels, called guard channels, exclusively for supporting handoff calls
- ❑ Unfortunately, guard channels increases blocking probability of new calls
- ❑ The number of guard channels is left to the operators to optimize (not part of the standard)

Cellular System Capacity Example

- A particular cellular system has the following characteristics: cluster size = 7, uniform cell size, user density = 100 users/sq km, allocated frequency spectrum = 900-949 MHz, bit rate required per user = 10 kbps uplink and 10 kbps downlink, and modulation code rate = 1 bps/Hz.
 - A. Using FDMA/FDD:
 - 1. How much bandwidth is available per cell using FDD?
 - 2. How many users per cell can be supported using FDMA?
 - 3. What is the cell area?
 - 4. What is the cell radius assuming circular cells?
 - B. If the available spectrum is divided into 35 channels and TDMA is employed within each channel:
 - 1. What is the bandwidth and data rate per channel?
 - 2. How many time slots are needed in a TDMA frame to support the required number of users?
 - 3. If the TDMA frame is 10ms, how long is each user slot in the frame?
 - 4. How many bits are transmitted in each time slot?

Cellular System Capacity (Cont)

- ❑ A particular cellular system has the following characteristics: cluster size = 7, uniform cell size, user density = 100 users/sq km, allocated frequency spectrum = 900-949 MHz, bit rate required per user = 10 kbps uplink and 10 kbps downlink, and modulation code rate = 1 bps/Hz.
- ❑ A. Using FDMA/FDD:
 1. How much bandwidth is available per cell using FDD?
 $49 \text{ MHz} / 7 = 7 \text{ MHz/cell}$
FDD $\Rightarrow$ 3.5 MHz/uplink or downlink
 2. How many users per cell can be supported using FDMA?
 $10 \text{ kbps/user} = 10 \text{ kHz} \Rightarrow 350 \text{ users per cell}$
 3. What is the cell area?
 $100 \text{ users/sq km} \Rightarrow 3.5 \text{ Sq km/cell}$
 4. What is the cell radius assuming circular cells?
 $\pi r^2 = 3.5 \Rightarrow r = 1.056 \text{ km}$

Cellular System Capacity (Cont)

B. If the available spectrum is divided in to 35 channels and TDMA is employed within each channel:

1. What is the bandwidth and data rate per channel?

$$3.5 \text{ MHz}/35 = 100 \text{ kHz/Channel} = 100 \text{ kbps}$$

2. How many time slots are needed in a TDMA frame to support the required number of users?

$$10 \text{ kbps/user} \Rightarrow 10 \text{ users/channel}$$

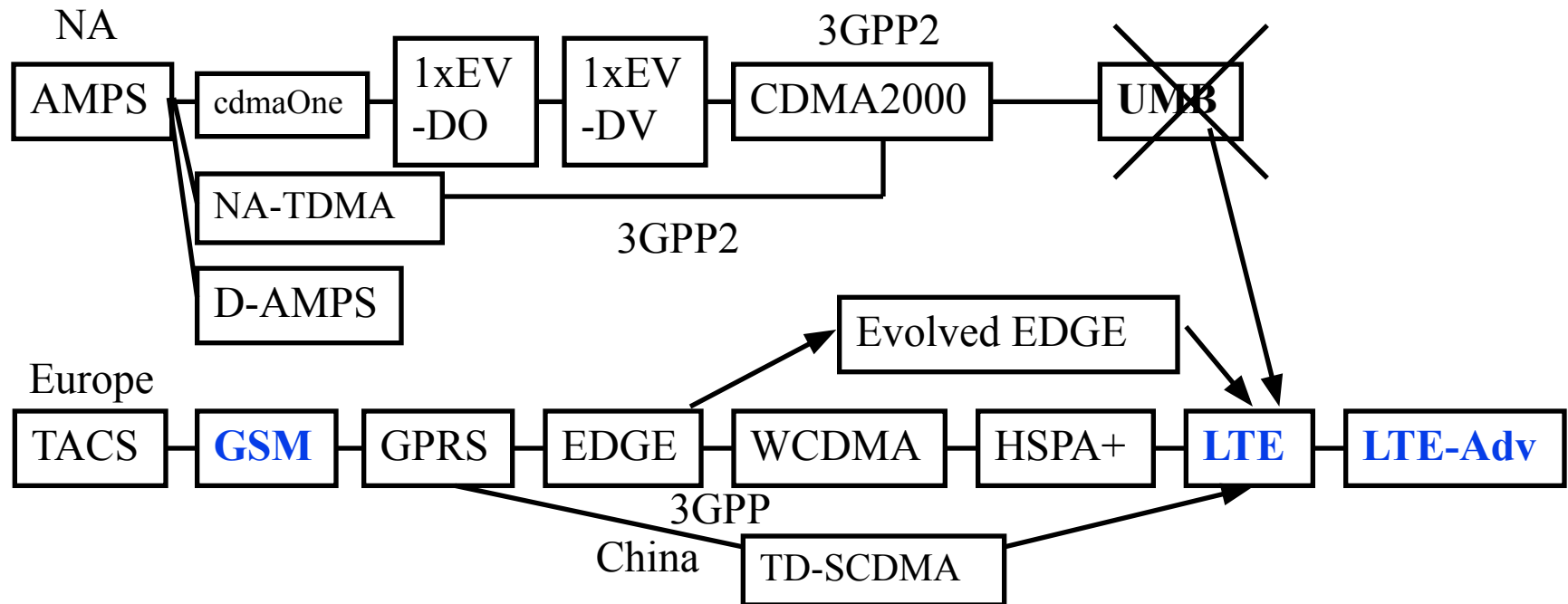
3. If the TDMA frame is 10ms, how long is each user slot in the frame?

$$10 \text{ ms}/10 = 1 \text{ ms}$$

4. How many bits are transmitted in each time slot?

$$1 \text{ ms} \times 100 \text{ kbps} = 100 \text{ b/slot}$$

Cellular Telephony Generations



Analog
FDMA

Digital
TDMA
CDMA

CDMA

OFDMA+ MIMO

Voice

Voice

Voice+Data

Voice+Data

Voice+HS Data

All-IP

1G 1983

2G 1990

2.5G

3G 2000

3.5G

4G 2013

Cellular Generations (Cont)

- ❑ **1G: Analog Voice. FDMA. 1980s**
 - ❑ AMPS: Advanced Mobile Phone System
 - ❑ TACS: Total Access Communications System
- ❑ **2G: Digital Voice. TDMA. 1990**
 - ❑ cdmaOne: Qualcomm. International Standard IS-95.
 - ❑ NA-TDMA
 - ❑ Digital AMPS (D-AMPS)
 - ❑ **GSM**: Global System for Mobile Communications
- ❑ **2.5G: Voice + Data. 1995.**
 - ❑ 1xEV-DO: Evolution Data Optimized
 - ❑ 1xEV-DV: Evolution Data and Voice
 - ❑ General Packet Radio Service (GPRS)
 - ❑ Enhanced Data Rate for GSM Evolution (EDGE)

Cellular Generations (Cont)

❑ **3G: Voice + High-speed data. All CDMA. 2000.**

- ❑ CDMA2000: Qualcomm. International Standard IS-2000.
- ❑ W-CDMA: Wideband CDMA
- ❑ TD-SCDMA: Time Division Synchronous Code Division Multiple Access (Chinese 3G)
- ❑ 384 kbps to 2 Mbps

❑ **3.5G: Voice + Higher-speed data**

- ❑ EDGE Evolution
- ❑ High-Speed Packet Access (HSPA)
- ❑ Evolved HSPA (HSPA+)
- ❑ Ultra Mobile Broadband (UMB)

Cellular Generations (Cont)

- ❑ Two Tracks for 1G/2G/3G:
 - ❑ Europe 3GPP (3rd Generation Partnership Project)
 - ❑ North America 3GPP2
- ❑ **3.9G: High-Speed Data. VOIP. OFDMA.**
 - ❑ Long Term Evolution (LTE)
- ❑ **4G: Very High-Speed Data. 2013.**
 - ❑ LTE-Advanced
 - ❑ 100 Mbps – 1 Gbps
- ❑ **5G: Ultra High-Speed Data. 2020.**
 - ❑ IP based

LTE/4G

LTE: Key Features

Long Term Evolution. 3GPP Release 8, 2009.

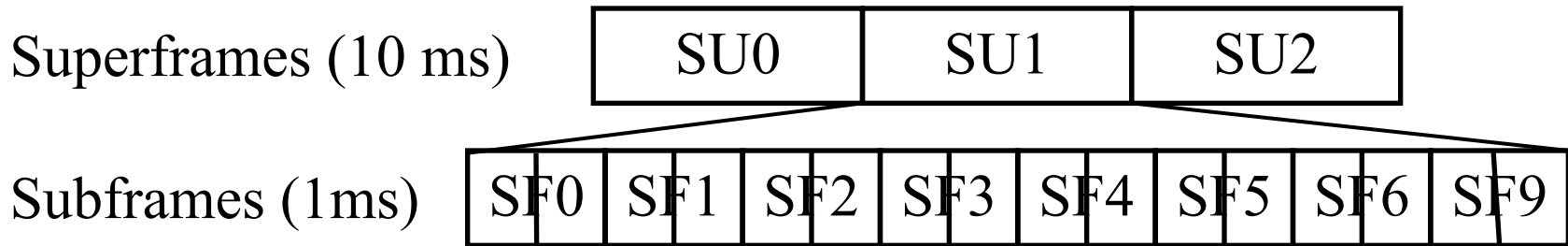
1. **Many different bands:** 700/1500/1700/**2100**/2600 MHz
2. **Flexible Bandwidth:** 1.4/3/5/10/15/20 MHz
3. Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD)
⇒ Both *paired* and *unpaired* spectrum
4. 4x4 MIMO, Multi-user collaborative MIMO
5. Beamforming in the downlink

Ref: A. Ghosh, J. Zhang, J. G. Andrews, R. Muhamed, "Fundamentals of LTE," Prentice Hall, 2010, ISBN: 0137033117, 464 pp. Safari book.

LTE: Key Features (Cont)

8. Data Rate: 326 Mbps/down 86 Mbps up (4x4 MIMO 20 MHz)
9. Modulation: OFDM with QPSK, 16 QAM, 64 QAM
10. **OFDMA** downlink,
Single Carrier Frequency Division Multiple Access
(**SC-FDMA**) uplink
11. **Hybrid ARQ** Transmission
12. Short **Frame Sizes** of 10ms and 1ms $\Rightarrow$ faster feedback and better efficiency at high speed
13. **Persistent scheduling** to reduce control channel overhead for low bit rate voice transmission.
14. **IP based** flat network architecture

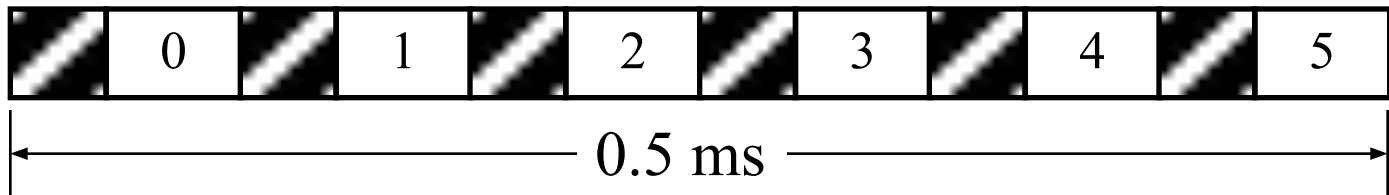
LTE Frame Structure



- ❑ Subframe = 2 slots of 0.5 ms each
- ❑ Slot = 6 or 7 symbols of $66.7 \mu\text{s}$ ($1/15 \text{ kHz}$) each
- ❑ Normal Cyclic Prefix: $5.2 \mu\text{s}$ for 1st symbol, $4.7 \mu\text{s}$ for others

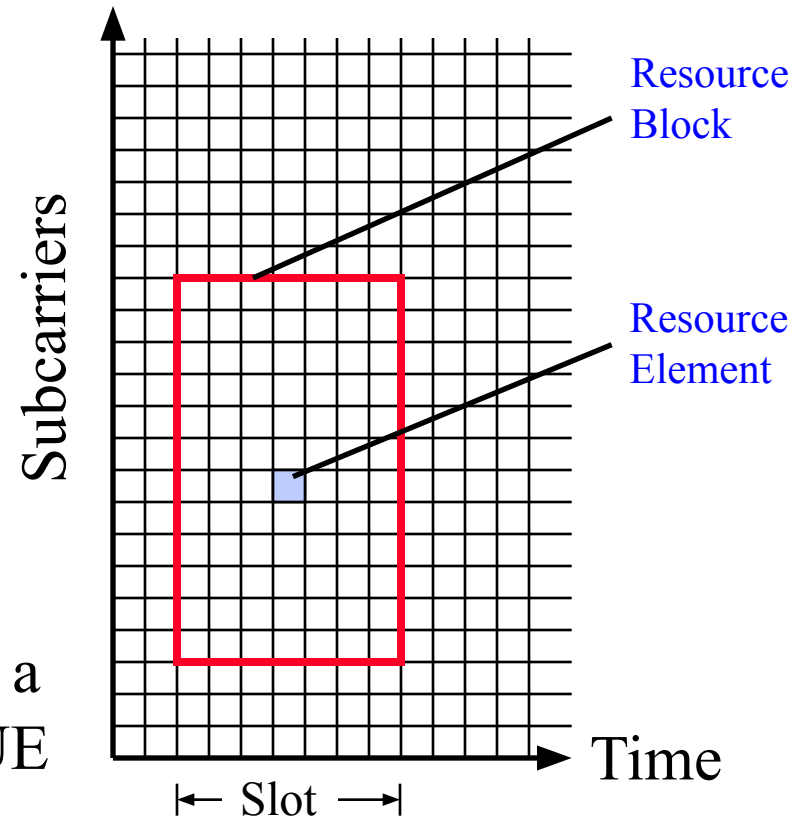
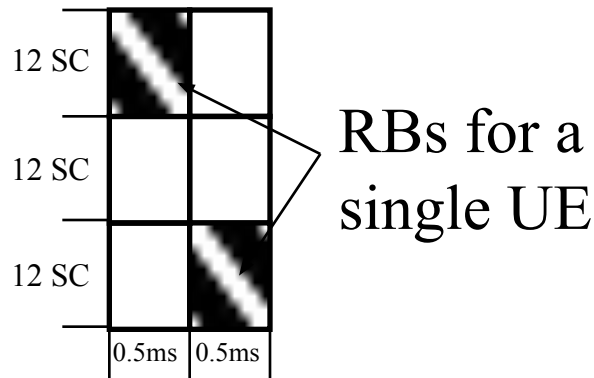


- ❑ Extended Cyclic Prefix: for larger networks. $16.7 \mu\text{s}$ each



Resource Allocation

- ❑ **Time slot:** 0.5 ms
6 or 7 OFDM symbols
- ❑ **Subcarriers:** 15 kHz
- ❑ **Physical Resource Block (RB):**
12 subcarriers (180 kHz)
over 1 time slot
- ❑ **Minimum Allocation:** 2 RBs
per subframe



Ref: A. Ghosh, J. Zhang, J. G. Andrews, R. Muhamed, "Fundamentals of LTE," Prentice Hall, 2010, ISBN: 0137033117, 464 pp.

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Example

❑ For *normal* cyclic prefix (CP), how many resource elements (REs) are there in 2 RBs?

❑ Solution

❑ With normal CP, we have 7 symbols per slot

❑ Number of REs per RB = $12 \times 7 = 84$

❑ Number of REs in 2 RB = $2 \times 84 = 168$

LTE Transmission Bandwidth

- ❑ For downlink, LTE does not use all subcarriers
- ❑ Transmission bandwidth < Channel bandwidth

Channel bandwidth [MHz]	1.4	3	5	10	15	20
Transmission bandwidth [MHz]	1.08	2.7	4.5	9	13.5	18
Transmission bandwidth [RB]	6	15	25	50	75	100

http://www.viavisolutions.com/sites/default/files/technical-library-files/LTE_PHY_Layer_Measurement_Guide_0.pdf

Example

- ❑ What is the *transmission bandwidth* for a resource allocation of 10 RBs?
- ❑ Solution
 - ❑ Each RB = 180 kHz
 - ❑ Transmission Bandwidth = $10 \times 180 = 1.8$ MHz

Example

- ❑ What is the *peak data rate* of DL LTE?
- ❑ Solution
 - ❑ For peak data rate, we assume best conditions, i.e., 64 QAM (6 bits per symbol), short CP (7 symbols per 0.5 ms slot), and 20 MHz channel
 - ❑ Each symbol duration = $0.5 \text{ ms} / 7 = 71.4 \mu\text{s}$
 - ❑ Number of RB for 20 MHz = 100
 - ❑ Number of subcarriers per RB = 12
 - ❑ Number of subcarriers for 20 MHz channel = $100 \times 12 = 1200$
 - ❑ Number of bits transmitted per symbol time = $6 \times 1200 \text{ bits}$
 - ❑ Data rate = $(6 \times 1200 \text{ bits}) / (71.4 \mu\text{s}) = 100.8 \text{ Mbps}$ (without MIMO)

5G

5G Promise

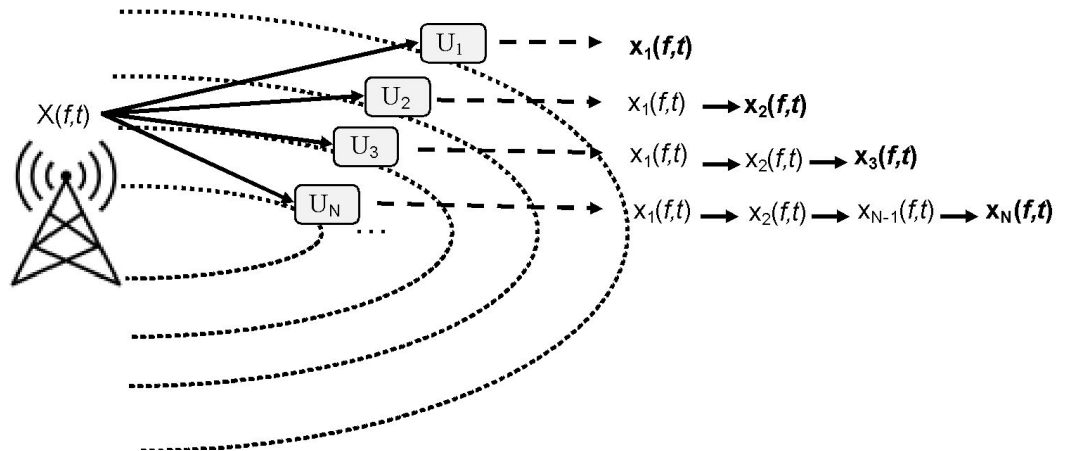
- ❑ Deployment started in 2019/2020
- ❑ Designed to improve not only the data rates, but many other things
- ❑ Key 5G targets
 - ❑ **1. Data Rate:** While 4G offered the maximum data rate of 1Gbps per user under ideal conditions, 5G promises **20Gbps** under the same conditions.
 - ❑ **2. Latency:** ~100ms with 3G and ~30ms in 4G. 5G promises **1ms**.
 - ❑ **3. Connection Density:** 4G could connect 100 thousand devices per km², 5G promises to connect **1 million/ km²**.
- ❑ 5G Applications
 - ❑ **Enhanced broadband:** fixed wireless (no cable/wire coming to homes), new videos standards (4K/8K, 360°), wireless VR, blazing photo/video upload, ...
 - ❑ **Ultra-reliable low latency communications:** autonomous driving, remote medical procedures, and so on.
 - ❑ **IoT:** will connect billions of devices at low energy, long distance, hard-to-reach areas

3 Fundamental Dimensions for Cellular Enhancements

- ❑ **Increase bps/Hz or spectral efficiency:** develop new coding and modulation techniques as well as new spectrum sharing methods to squeeze more bits out of the given spectrum. Increases capacity linearly.
- ❑ **Reduce cell radius or increase spectral reuse:** Smaller cells allow higher spectrum reuse in the service area. The most effective method to increase capacity. Cell sizes have been consistently reduced over the 4 generations. 5G will continue to follow this trend.
- ❑ **Use new spectrum:** Eventually we will need new spectrum to cope with the increasing demand for mobile traffic. 5G will be the first generation to use millimeter wave bands.

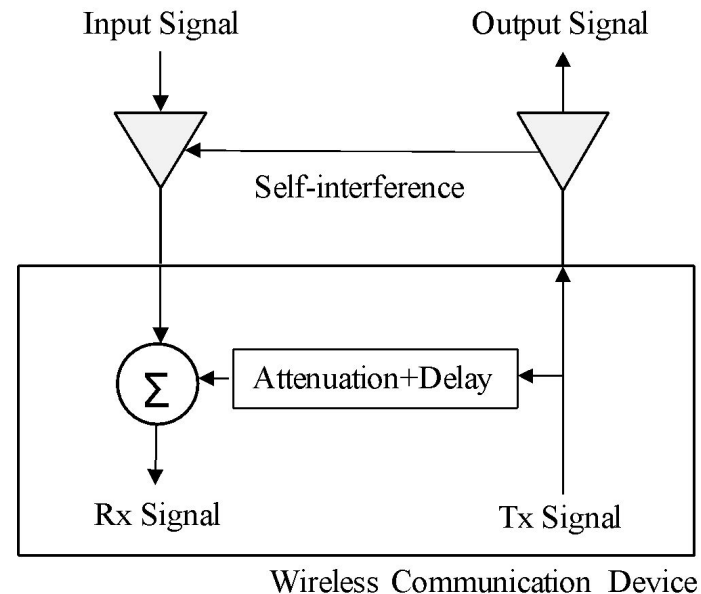
NOMA

- Use power as the 4th dimension of multiplexing
- Allows use of the same frequency at the same time for all users
- BS transmits combined signal with the highest power for the farthest user's signal
- Devices decode the highest –power signal first by treating all other as noise; then removes it from the combined signal; stops when own signal is received (successive interference cancellation or SIC)



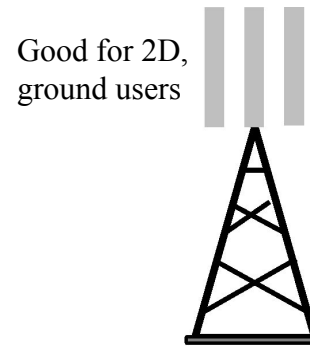
Full-duplex Wireless

- ❑ Full-duplex in wireless has not been possible so far due to self-interference
- ❑ Half-duplex reduces capacity and increases latency
- ❑ With advanced processing, attenuation+delay circuits within the radio hardware can cancel self-interference and achieve full-duplex

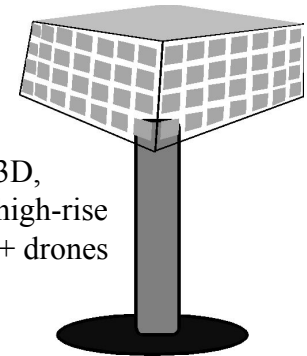


Massive MIMO and 3D Beamforming

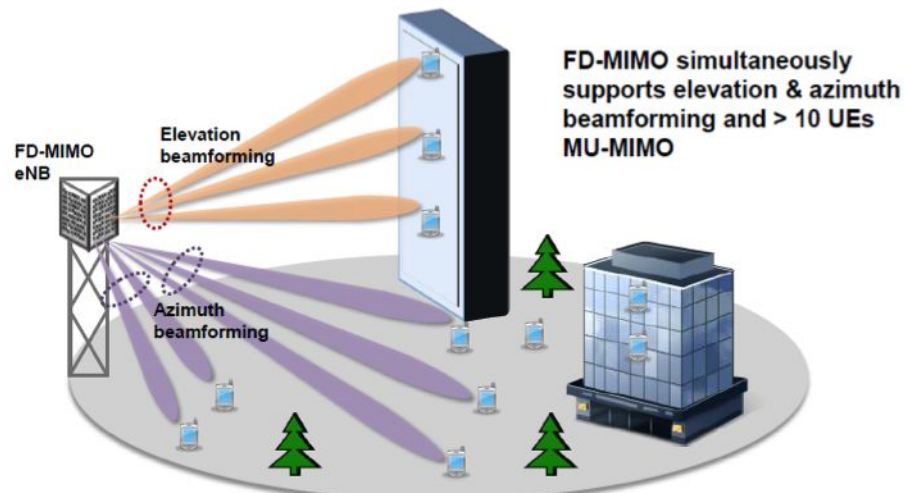
- Existing BSs use vertical antennas; good for serving ground users
- 5G BS will have planar array antennas with many (>100) antenna elements; 3D beams formed by adjusting phase and amplitude of each antenna element



4G Sector Antenna

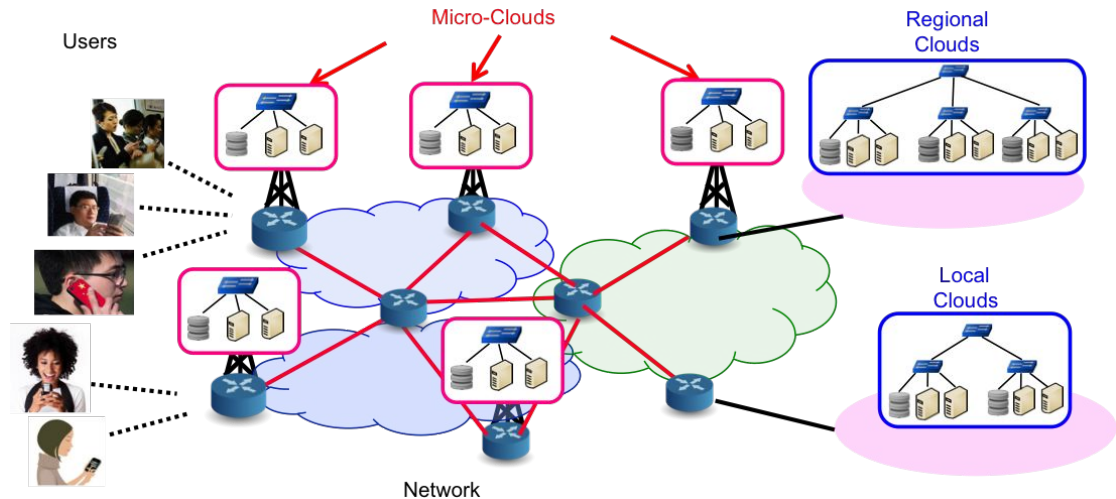


5G Massive MIMO



Mobile Edge Computing

- Future handsets will need many computations not feasible within the device (e.g., natural language processing, augmented reality, etc.); cloud computing will increase latency
- Provision mini-clouds in each radio tower (at the edge) to provide computing power with low latency and energy cost



Summary

- ❑ In a cellular cluster of size N , the minimum distance between cells with same frequencies is $D = R \sqrt{3N}$. Here R is the cell radius.
- ❑ 1G was analog voice with FDMA
- ❑ 2G was digital voice with TDMA. Most widely implemented 2G is GSM. Data rate was improved by GPRS and EDGE.
- ❑ 3G was voice+data with CDMA. Most widely implemented 3G is W-CDMA using two 5 MHz FDD channels. Data rate was improved later using HSPA and HSPA+.
- ❑ LTE uses a **super-frame** of 10 subframes of 1 ms each. Each **subframe** has one 0.5 ms **slot** for uplink and downlink each.
- ❑ 5G is being launched in 2020 promising to offer ultra-high data rates, ultralow latency, and massive connectivity for Internet of Things
- ❑ 5G will use NOMA as a new access technology that enables serving multiple users over the same frequency at the same time; NOMA uses power as a new dimension to differentiate users.
- ❑ 5G promises full-duplex wireless communications where both the Tx and Rx antennas can function at the same time.
- ❑ 5G base stations will use planar array antennas for massive MIMO and 3D beamforming.
- ❑ 5G base stations will host computing and storage resources to reduce latency for applications requiring cloud support.
- ❑ 5G will use new spectrum in the mmWave band.